Fisheries strategies and climate changes synergetically impact the Western Baltics species distribution, spread, and biomass, being the **fishery strategy the leading factor of the changes**

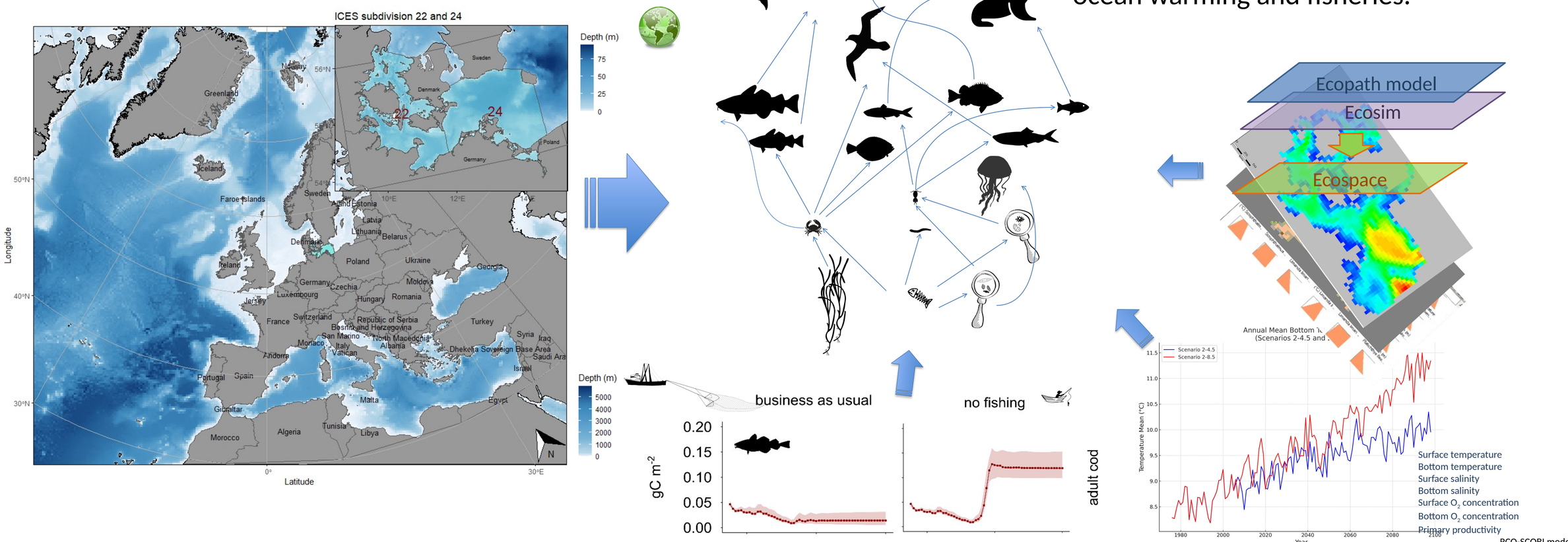
Multiple stressors in the future Western Baltic Sea: A climate-driven ecosystem model

Nascimento M.C.[1] and Scotti, M[1]

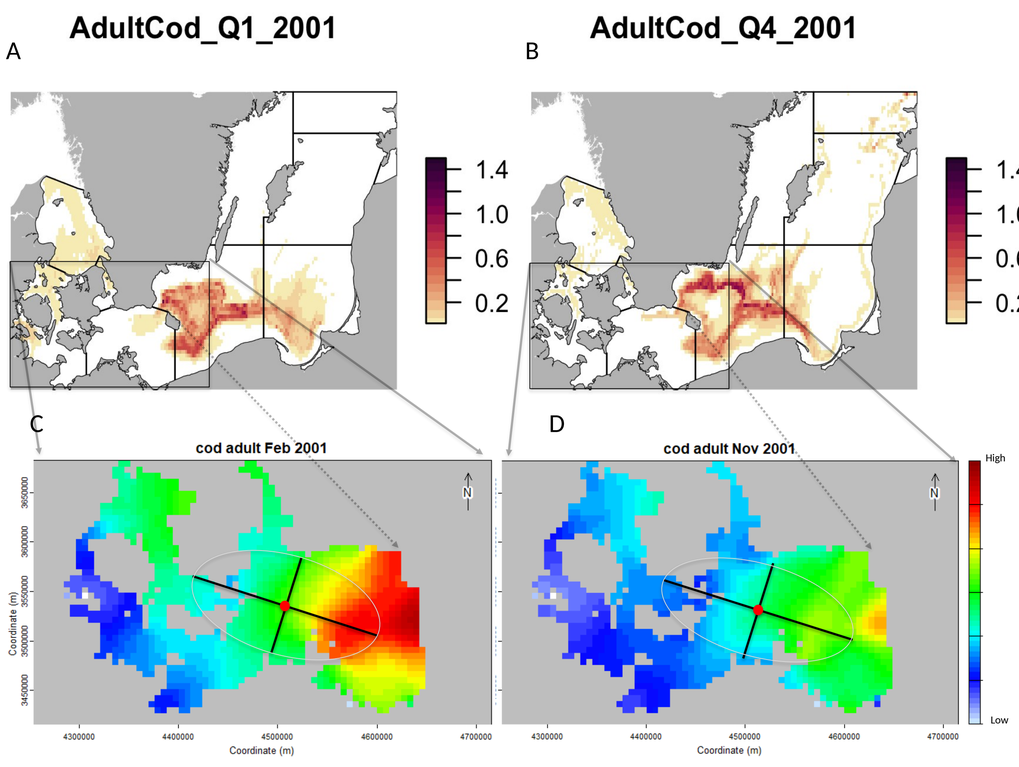
[1] Marine Evolutionary Ecology – RD Marine Ecology, Helmholtz Centre for Ocean Kiel - GEOMAR, Kiel, Germany

Ecopath with Ecosim model, with 18 functional groups (FG) - Western Baltic Sea (WBS) ecosystem, ICES areas 22 and 24, in 1994 (Scotti et al., 2022).

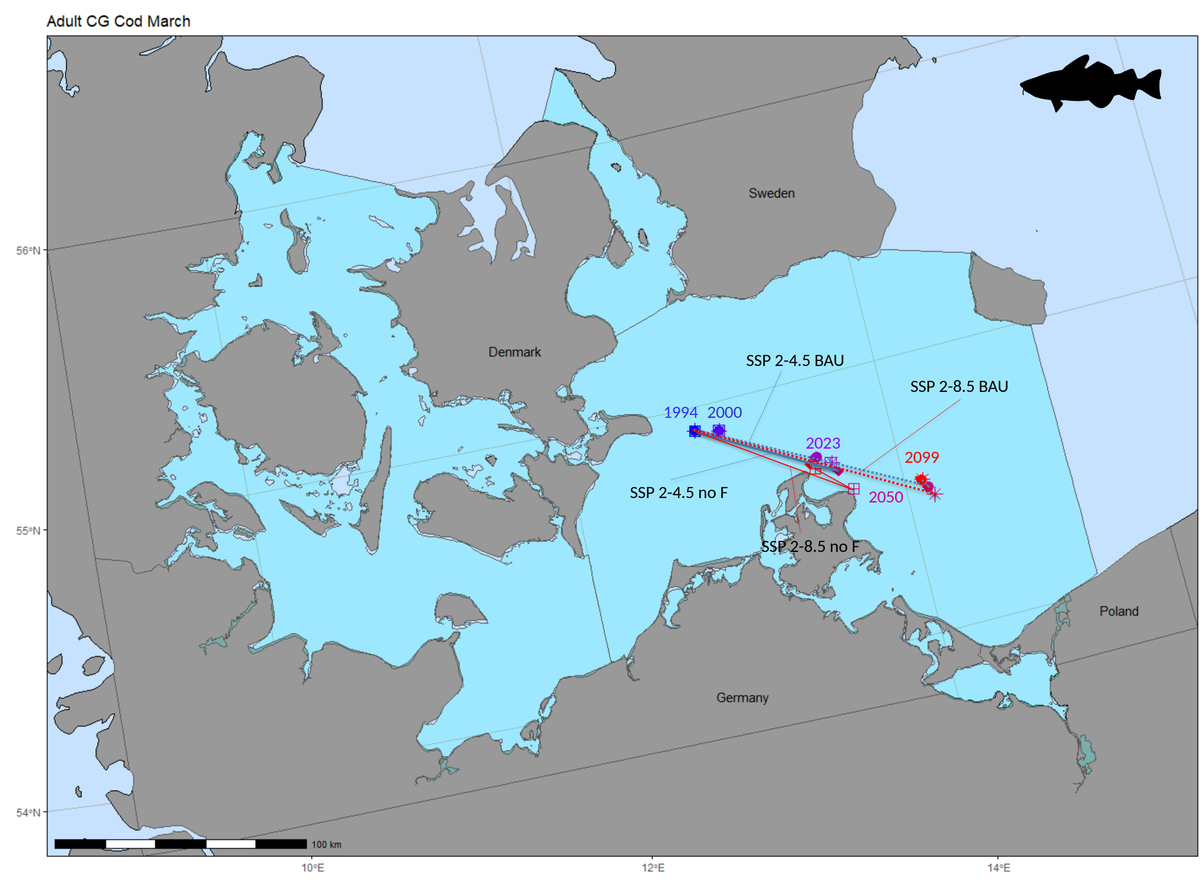
Using Ecospace's time-dynamic framework, we simulated four scenarios using the outcome of a biogeochemical model and multiple stressors such as ocean warming and fisheries.



We compared the Ecospace distribution with the observed distribution to validate the model. We post-processed the Ecospace outputs, calculating the centre of gravity (CG) and inertia of all maps generated by time steps. Then we compared the scenarios based on the CG, the Inertia and the biomass ratio.



A and **B** Western Baltic Adult Cod density distribution calculated using an individual distribution model based on ICES data by **Liam MacNeil** for quarters 1 and 4. **C** and **D** Western Baltic Adult Cod biomass distribution in February and November of 2001, estimated by the Ecospace model. The cross represents inertia, and the red point in the centre of the map represents the centre of gravity.



Climate change shifts the distribution of functional groups in the western Baltic Sea ecosystem by also changing their biomass. It influences changes in the distribution of important fish stocks. However, fisheries are the most important factor of change for the main commercial targets, such as the western Baltic cod.

	Scenario 2 – 4.5				Scenario 2 – 8.5			
	BAU		No F		BAU		No F	
Time interval	94-50	50-99	94-50	50-99	94-50	50-99	94-50	50-99
Distance shift CG (km)	96	4	60	12	99	8	68	14
Inertia ratio	0.65	1.06	0.65	1.16	0.33	1.11	0.62	1.22
Biomass ratio (E/S)	1.35e-6	(94-2099) 2.66e-3	4.09e+1	(94-2099) 5.72e+1	7.96e-7	(94-2099) 0.009	3.93e+1	(94-2099) 6.24e+1

GEFÖRDERT VOM
DAM DEUTSCHE ALLIANZ MEERESFORSCHUNG
Bundesministerium für Bildung und Forschung
Freie Hansestadt Bremen
Hamburg Behörde für Wissenschaft, Forschung, Gleichstellung und Bezirke
Niedersächsisches Ministerium für Wissenschaft und Kultur
MV Mecklenburg-Vorpommern
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